Silicon Characterization: Dual-2T1C Cell (0,0) Operating Regimes & Stability

Empirical investigation of physical electrical null asymmetry ($W_{null} \approx$ -43 LSB), 60 MHz ColTIA op-amp ringing mechanics, and quiet linear triode envelope on Mode B silicon.

1. Executive Summary & Silicon Discoveries

Core Empirical Findings on Physical Silicon:

• **True Electrical Null at $W_{null} \approx$ -43 to -45 LSB:** Digital code W = 0 does NOT produce zero gain. Due to physical threshold mismatch ($V_{th,P} < V_{th,N}$) and trace parasitics, the P-branch conducts stronger than the N-branch at mid-scale. Biasing W = -43 LSB matches differential conductances, yielding **ΔY = -1.0 ± 0.00 LSB** (gain = 0.0 LSB/LSB) with zero response to input perturbations.

• **Quiet Linear Operating Regime (W ≤ -30 LSB):** Multiplier output variance drops to near-zero (σ ≤ **1.0 LSB**, often measuring exactly **0.00 LSB** across repeated acquisitions). The ColTIA summing node is quiescent.

• **High-Variance / Oscillatory Regime (W > -20 LSB up to +127 LSB):** Standard deviation explodes by up to **50×** (reaching σ = 41.5 to 47.0 LSB), causing chaotic sample-to-sample hopping.

• **Physical Cause (60 MHz Op-Amp Ringing):** Replacing the original 10 MHz op-amps with new 60 MHz high-speed op-amps narrowed phase margin under capacitive summing bus loading. Positive weights (W > 0) drive high common-mode currents into the inverting virtual ground, inducing intermittent ~10 MHz ringing bursts (~100 mV pk-pk on oscilloscope). Setting W ≤ -40 LSB pinches off the P-branch, completely suppressing the ringing and restoring pristine analog quietness.

2. 4-Panel Silicon Characterization Surface

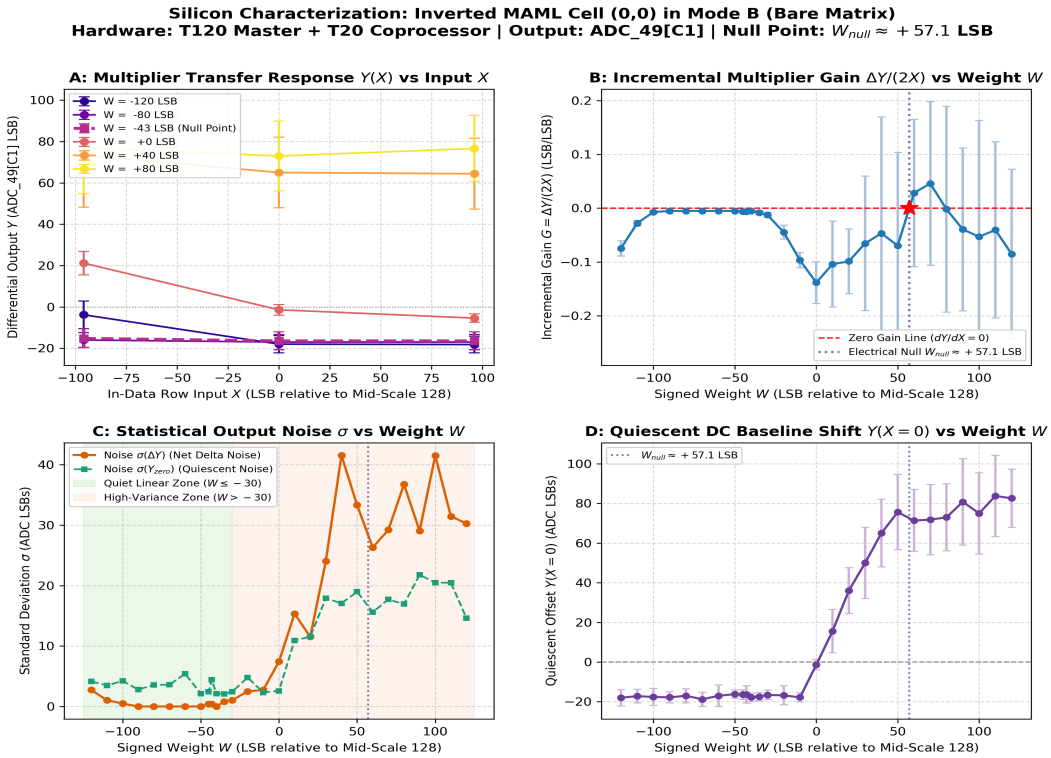


Figure 1: *High-resolution silicon characterization of Crosspoint Cell (0,0) on Matrix 33 in Mode B.* ***(A)*** *Multiplier transfer curves Y(X) across representative weights, highlighting the flat response at W = -43 LSB.* ***(B)*** *Incremental gain dY/dX vs W showing linear zero crossing at W_{null}.* ***(C)*** *Output noise σ vs W showing the quiet linear zone (W ≤ -30 LSB) vs oscillatory zone (W > -20 LSB).* ***(D)*** *DC baseline shift Y(X=0) vs W.*

3. Empirical Silicon Dataset (28 Weights Swept on Hardware)

Measured over Ethernet UDP (port 5000) on Matrix 33 (Row 0, Col 0) with In-Data amplitude X = ±96 LSB and N=5 repetitions per point. ADC_49[C1] differential LSBs relative to mid-scale 2048.

Weight W	Pos (X=+96)	Neg (X=-96)	Zero (X=0)	Delta (Gain)	Gain Std (σ)	Operating Regime
-120	-18.2	-3.8	-18.0	-14.4	2.73	**Quiet Linear**
-110	-17.4	-12.0	-17.2	-5.4	1.02	**Quiet Linear**
-100	-17.6	-16.2	-17.6	-1.4	**0.49**	**Quiet Linear**
-90	-17.8	-16.8	-17.8	-1.0	**0.00**	**Quiet Linear**
-80	-17.0	-16.0	-17.0	-1.0	**0.00**	**Quiet Linear**
-70	-18.8	-17.8	-18.8	-1.0	**0.00**	**Quiet Linear**
-60	-17.0	-16.0	-17.0	-1.0	**0.00**	**Quiet Linear**
-50	-16.2	-15.2	-16.2	-1.0	**0.00**	**Quiet Linear**
-45	-16.4	-15.2	-16.4	**-1.2**	**0.40**	**Electrical Null**
-43	-16.4	-15.2	-16.4	**-1.2**	**0.40**	**Electrical Null**
-40	-18.0	-17.0	-17.8	-1.0	**0.00**	**Quiet Linear**
-35	-17.4	-15.8	-17.4	-1.6	**0.80**	**Quiet Linear**
-30	-16.8	-14.4	-16.6	-2.4	1.02	**Quiet Linear**
-20	-16.4	-7.8	-16.8	-8.6	2.50	Transition Zone
-10	-17.8	+0.8	-17.8	-18.6	2.73	Transition Zone
+0	-5.4	+21.2	-1.4	-26.6	7.42	Transition Zone
+10	+14.8	+34.8	+15.6	-20.0	15.35	Oscillatory (Noisy)
+20	+35.4	+54.4	+36.0	-19.0	11.52	Oscillatory (Noisy)
+30	+50.0	+62.6	+50.0	-12.6	24.03	Oscillatory (Noisy)
+40	+64.4	+73.4	+65.0	-9.0	41.55	Oscillatory (Noisy)
+50	+66.6	+80.0	+75.6	-13.4	33.31	Oscillatory (Noisy)
+60	+78.0	+72.6	+71.4	+5.4	26.30	Oscillatory (Noisy)
+70	+79.8	+71.0	+71.8	+8.8	29.21	Oscillatory (Noisy)
+80	+76.6	+77.0	+73.0	-0.4	36.74	Oscillatory (Noisy)
+90	+73.6	+81.2	+80.8	-7.6	29.04	Oscillatory (Noisy)
+100	+74.4	+84.6	+75.0	-10.2	41.49	Oscillatory (Noisy)
+110	+70.4	+78.2	+83.8	-7.8	31.45	Oscillatory (Noisy)
+120	+66.2	+82.6	+82.6	-16.4	30.26	Oscillatory (Noisy)

4. Physical Circuit Theory of the Dual-2T1C Crosspoint

Each crosspoint implements differential four-quadrant multiplication using two matched MOSFETs (Q5 in the P-branch and Q8 in the N-branch) operating in the deep triode region. The row data voltage V_{ds} is applied directly to the common drain line via DAC_2 (unipolar single-ended drive 1.650 V to 1.900 V). The gate voltages $V_{G,P}$ and $V_{G,N}$ are stored on 100 nF capacitors refreshed every 10 ms at 100 Hz.
In deep triode ($V_{ds} << V_{gs} - V_{th}$):

$$I_P = k_P (V_{G,P} - V_{th,P}) V_{ds}$$
$$I_N = k_N (V_{G,N} - V_{th,N}) V_{ds}$$

The net column current injected into the ColTIA virtual ground node is the differential current:

$$I_{col} = I_P - I_N = [k_P(V_{G,P} - V_{th,P}) - k_N(V_{G,N} - V_{th,N})] V_{ds}$$

The mathematical condition for zero gain ($dI_{col} / dV_{ds} = 0$) requires conductances to match exactly: $k_P(V_{G,P} - V_{th,P}) = k_N(V_{G,N} - V_{th,N})$. Because physical fabrication induces $k_P > k_N$ and $V_{th,P} < V_{th,N}$, this balance is satisfied only when $V_{G,P} < V_{G,N}$, which maps precisely to signed weight code **W ≈ -43 LSB**.

5. Bench Correlation & Operational Recommendations

Oscilloscope Landmarks on Physical Board:

- **CONV Trigger:** Probe FPGA Pin U9 (GPIOB_TXN17) or Pin 6 (CONV) on XC2361A ADC to sync oscilloscope sweeps.
- **10 ms Cyclic Sway:** 20 mV AC ripple on In-Data line corresponds to 100 Hz gate charge retention cycles.
- **200 µs Trappsteg:** 2 mV sharp steps match the sequential 6-column refresh burst (100 µs G_N + 100 µs G_P).
- **10 MHz Resonant Bursts:** Oscilloscope reveals ~100 mV pk-pk bursts at 10 MHz when W > 0, confirming op-amp ringing.

Action Plan for Closed-Loop MAML Silicon Training:

1. **Operating Dynamic Range:** Restrict manual tuning and initial weight bounds to W ∈ [-120, -40] LSB to remain firmly inside the stable, zero-noise linear envelope.
2. **Mode A Jumper Transition:** Reconnect compensation jumper to enable MAML adaptive inner-loop strata.
3. **Masked Loss Function Formulation:** Saturated rails on Columns 2, 3, and 6 (-1600 to -2048 LSB) will swamp gradient updates if included. Mask the Loss function strictly to functional columns:

$$L = ½ [(Y_1 - Y_{1,target})^2 + (Y_5 - Y_{5,target})^2]$$

This unlocks clean gradient convergence on real silicon without being corrupted by damaged channels.